Type MS · Using a line's equation to find a missing coordinate or intercept

Answer 1 of 7 · Question 9 part a · 2 marks

2020 JAN · Paper 01 (regional R) · extract 1 of 3 · find part a in this mark scheme extract

Number		
9 (a)	$a=-1,\ b=-2$	B1,B1 [2]
(b)	Gradient of $l_1=-2,\Rightarrow$ Gradient of $l_2=\frac{1}{2}$	B1,B1
	$180=(x+1)^2+(y-6)^2$	M1
	$\frac{1}{2}=\frac{y-6}{x+1}\Rightarrow x=2y-13$	M1
	Solves simultaneous equations; $180=([2y-13]+1)^2+(y-6)^2\Rightarrow 0=5y^2-60y$	M1M1
	or $180=(x+1)^2+\left(\frac{1}{2}x+\frac{13}{2}-6\right)^2\Rightarrow 0=x^2+2x-143=0$	
	$y=0,\ y=12\Rightarrow x=-13,\ x=11$ or $x=-13,\ x=11\Rightarrow y=0,\ y=12$ Coordinates are $(-13,\ 0)$ and $(11,\ 12)$	A1A1 [8]
(c)	Area of triangle PQR $PQ=\sqrt{(6+2)^2+(-1-3)^2}=4\sqrt{5}$	M1
	Area $=\frac{1}{2}\times 4\sqrt{5}\times 6\sqrt{5}=60$ (units)^2	M1A1 [3]
	ALT Area $=\frac{1}{2}\begin{pmatrix}-13 & -1 & 3 & -13\\ 0 & 6 & -2 & 0\end{pmatrix}=\frac{1}{2}(-78+2+0-0-18-26)=-60$	{M1}{M1}
	$\Rightarrow 60$ (units)^2	{A1} [3]
(d)	Coordinates of R required are $(-13,\ 0)$	
	$\angle RPQ=90^\circ$ so RQ is a diameter	
	$\left(\frac{-13+3}{2},\frac{0-2}{2}\right)\Rightarrow(-5,\ -1)$	M1A1 [2]
		Total 15 marks

Type MS · Using a line's equation to find a missing coordinate or intercept

Answer 1 of 7 · Question 9 part a · 2 marks

2020 JAN · Paper 01 (regional R) · extract 2 of 3 · find part a in this mark scheme extract

(b)	
B1	Gradient of $l_1 = -2$
B1	Gradient of $l_2 = \frac{1}{2}$
M1	Use of $PR = 6\sqrt{5}$ to obtain an equation
M1	Use of gradient of the perpendicular to obtain an equation
M1	Solves simultaneously
M1	Simplifies to $5y^2 - 60y = 0$ or $x^2 + 2x - 143 = 0$
A1	All 4 values identified 0, 12, –13, 11
A1	$(11,\ 12)$ and $(-13,\ 0)$ (must be paired correctly and if written as a coordinate then must be in the correct order)
(c)	
M1	$PQ = 4\sqrt{5}$
M1	Use of Area $= \frac{1}{2} \times PQ \times PR$
A1	60 (units)^2
ALT	
M1	Use of Area $= \frac{1}{2}\begin{pmatrix} -13 & -1 & 3 & -13 \\ 0 & 6 & -2 & 0 \end{pmatrix}$ ft R provided $e < 0$
M1	$\frac{1}{2}(-78 + 2 + 0 - 0 - 18 - 26)$ ft R provided $e < 0$
A1	60 (units)^2
(d)	
M1	$\left(\frac{-13+3}{2}, \frac{0-2}{2}\right)$ ft R provided $e < 0$
A1	$(-5,\ -1)$

Type MS · Using a line's equation to find a missing coordinate or intercept

Answer 1 of 7 · Question 9 part a · 2 marks

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Question number	Scheme	Marks

Type MS · Using a line's equation to find a missing coordinate or intercept

Answer 2 of 7 · Question 10 part a · 1 mark

2022 JAN · Paper 01 · extract 1 of 4 · find part a in this mark scheme extract

10 (a)	For the correct value of $a = 10$	B1 [1]
(b)	Gradient of line L_2 $m = -\frac{1}{2}$ $y - '10' = '-\frac{1}{2}'(x-2)$ $y - 10 = -\frac{1}{2}(x-2)$ oe $x + 2y - 22 = 0$ *	B1 M1 A1 A1 cso [4]
(c)	Coordinates of point A are $(-3,\ 0)$ Coordinates of point B are $(22,\ 0)$ Length of AC $(5\sqrt{2})^2 = (m - '-3')^2 + n^2$ $[\Rightarrow 50 = m^2 + 6m + 9 + n^2]$ Gradient of BC $\frac{1}{4} = \frac{n}{m-'22'} \Rightarrow n = \frac{m-'22'}{4} \Rightarrow n^2 = \frac{m^2-44m+484}{16}$ $50 = m^2 + 6m + 9 + \frac{m^2-44m+484}{16}$ [or $50 = (22+4n)^2 + 6(22+4n) + 9 + n^2$] $17m^2 + 52m - 172 = 0$ OR $17n^2 + 200n + 575 = 0$ e.g. $m = \frac{-52 \pm \sqrt{52^2 - 4\times17\times-172}}{2\times17} \Rightarrow m = 2,\ (-5.058...)$ $m = 2$ and $n = -5$	B1 B1 M1 M1 ddM1 A1 M1 A1A1 [9]
(d)	Area$_{APB} = \frac{1}{2}('10') \times ('22' - '3') = (125)$ Area$_{ABC} = \frac{1}{2}('5') \times ('22' - '-3') = \left(\frac{125}{2}\right)$ Area of quadrilateral $ACBP = \frac{375}{2}$ (units)^2	M1 A1 A1 [3]
	ALT	
	Area$_{ACBP} = \frac{1}{2}('25') \times ('15') = ...$ Area of quadrilateral $ACBP = \frac{375}{2}$ (units)^2	[M1A1 A1]
	ALT	

Type MS · Using a line's equation to find a missing coordinate or intercept

Answer 2 of 7 · Question 10 part a · 1 mark

2022 JAN · Paper 01 · extract 2 of 4 · find part a in this mark scheme extract

	$A=\frac{1}{2}\begin{bmatrix}\text{'}-3\text{'} & 2 & \text{'}22\text{'} & \text{'}2\text{'} & \text{'}-3\text{'}\\ 0 & \text{'}10\text{'} & 0 & \text{'}-5\text{'} & \text{'}0\text{'}\end{bmatrix}$	[M1
	$A=\frac{1}{2}\left(\left[(-3)\times 10+2\times 0+22\times(-5)+2\times 0\right]-\left[2\times 0+22\times 10+2\times 0+(-3)\times -5\right]\right)=\ldots$	A1
	Area of quadrilateral $APBC = \frac{375}{2}$ (units)^2	A1]
		Total 17 marks

Type MS · Using a line's equation to find a missing coordinate or intercept

Answer 2 of 7 · Question 10 part a · 1 mark

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		Any other values of m and n must be rejected.
(d)	**M1**	For either area of triangle APB or ABC $\text{Area}_{APB} = \frac{1}{2}('10')\times('22'--'3') = (125)$ or $\text{Area}_{ABC} = \frac{1}{2}('5')\times('22'-'-3') = \left(\frac{125}{2}\right)$
	A1	Either area of triangle APB or ABC correct
	A1	Area of quadrilateral $ACBP = \frac{375}{2}$ (units)^2
ALT		
	M1A1	$\text{Area}_{ACBP} = \frac{1}{2}('25')\times('15') = \ldots$
	A1	Area of quadrilateral $ACBP = \frac{375}{2}$ (units)^2
ALT – determinant method		
	M1	For using the 'determinant method' e.g. $A = \frac{1}{2}\begin{bmatrix} '-3' & 2 & '22' & '2' & '-3' \\ 0 & '10' & 0 & '-5' & '0' \end{bmatrix}$
	A1	For a correct evaluation of their determinants using **their** values e.g. $A = \frac{1}{2}\left(\left[(-3)\times10+2\times0+22\times(-5)+2\times0\right]-\left[2\times0+22\times10+2\times0+(-3)\times-5\right]\right) = \ldots$
	A1	Area of quadrilateral $APBC = \frac{375}{2}$ (units)^2

Type MS · Using a line's equation to find a missing coordinate or intercept

Answer 2 of 7 · Question 10 part a · 1 mark

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Question	Scheme	Marks

Type MS · Using a line's equation to find a missing coordinate or intercept

Answer 3 of 7 · Question 10 part a · 2 marks

2025 NOV · Paper 01 · extract 1 of 4 · find part a in this mark scheme extract

Question	Scheme	Marks
10 (a)	$10-3a+a=0 \Rightarrow 10=2a \Rightarrow a=5$ *	M1A1cso [2]
(b)	$(0,-5)$	B1 [1]
(c) (i)	$A \quad (5,10)$ $C \quad (50,-5)$ $y=3x-5 \Rightarrow m_{L1}=3 \qquad m_{L2}=\dfrac{-5-10}{50-5}=\dfrac{-1}{3}$ $m_{L2}\times m_{L1}=\dfrac{-1}{3}\times 3=-1$	B1 M1A1 [3]
(ii)	$y-(-5)=\dfrac{-1}{3}(x-50) \Rightarrow y=\dfrac{-x}{3}+\dfrac{35}{3}$	M1A1 [2]
(d)	$-3=\dfrac{n-("-5")}{m} \Rightarrow n=-5-3m$ $2250=(50-m)^2+(-5-n)^2$ $2250=(50-m)^2+(-5-(-5-3m))^2=(50-m)^2+(3m)^2$ $2250=2500-100m+10m^2 \Rightarrow 0=25-10m+m^2=(m-5)^2$ $\Rightarrow m=5$ $\Rightarrow n=-5-3\times 5=-20$ **ALT** $-3=\dfrac{n-("-5")}{m} \Rightarrow m=\dfrac{-(n+5)}{3}$ $2250=(50-m)^2+(-5-n)^2$ $2250=\left(50+\dfrac{n+5}{3}\right)^2+n^2+10n+25$ $2250=2500+\dfrac{100(n+5)}{3}+\dfrac{n^2+10n+25}{9}+n^2+10n+25$ $\Rightarrow \quad 0=9(2500-2250)+300(n+5)+10n^2+100n+250$ $=2500+300n+1500+10n^2+100n$ $=n^2+40n+400$ $=(n+20)(n+20)$ $\Rightarrow n=-20$ $\Rightarrow m=\dfrac{-(-20+5)}{3}=5$	M1 M1 M1 M1 A1 A1 [6] [M1] [M1] [M1] [M1] [A1] [A1]
(e)	Correctly uses the formula for a kite using their lengths	M1

Type MS · Using a line's equation to find a missing coordinate or intercept

Answer 3 of 7 · Question 10 part a · 2 marks

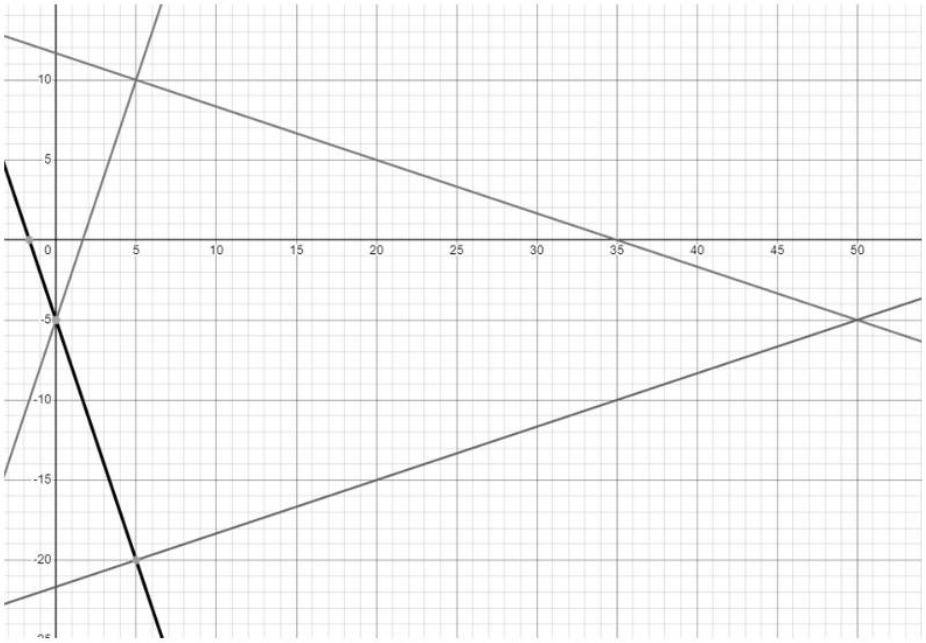
2025 NOV · Paper 01 · extract 2 of 4 · find part a in this mark scheme extract

	$\text{Area} = \frac{50\times(10-(-20))}{2} = 750$	dM1A1 [3]
	ALT 1 Correctly uses determinant method with their co-ordinates $\text{Area} = \frac{1}{2}\begin{pmatrix} 0 & 5 & 50 & 5 & 0 \\ -5 & 10 & -5 & -20 & -5 \end{pmatrix} = 750$	[M1 dM1A1]
	ALT 2 Correctly uses formulae for two triangles ABD and ACD using their lengths $\text{Area} = 2\times\left(\frac{1}{2}\times 15\times 50\right) = 750$ Or $\text{Area} = \left(\frac{1}{2}\times 30\times 5\right)+\left(\frac{1}{2}\times 45\times 30\right) = 750$	[M1 dM1A1]
		Total 17 marks

Type MS · Using a line's equation to find a missing coordinate or intercept

Answer 3 of 7 · Question 10 part a · 2 marks

2025 NOV · Paper 01 · extract 3 of 4 · find part a in this mark scheme extract



Type MS · Using a line's equation to find a missing coordinate or intercept

Answer 3 of 7 · Question 10 part a · 2 marks

2025 NOV · Paper 01 · extract 4 of 4 · find part a in this mark scheme extract

Question	Scheme	Marks

Type MS · Using a line's equation to find a missing coordinate or intercept

Answer 4 of 7 · Question 10 part b · 1 mark

2025 NOV · Paper 01 · extract 1 of 4 · find part b in this mark scheme extract

Question	Scheme	Marks
10 (a)	$10-3a+a=0 \Rightarrow 10=2a \Rightarrow a=5$ *	M1A1cso [2]
(b)	$(0,-5)$	B1 [1]
(c) (i)	$A \quad (5,10)$ $C \quad (50,-5)$ $y=3x-5 \Rightarrow m_{L1}=3 \qquad m_{L2}=\dfrac{-5-10}{50-5}=\dfrac{-1}{3}$ $m_{L2}\times m_{L1}=\dfrac{-1}{3}\times 3=-1$	B1 M1A1 [3]
(ii)	$y-(-5)=\dfrac{-1}{3}(x-50) \Rightarrow y=\dfrac{-x}{3}+\dfrac{35}{3}$	M1A1 [2]
(d)	$-3=\dfrac{n-("-5")}{m} \Rightarrow n=-5-3m$ $2250=(50-m)^2+(-5-n)^2$ $2250=(50-m)^2+(-5-(-5-3m))^2=(50-m)^2+(3m)^2$ $2250=2500-100m+10m^2 \Rightarrow 0=25-10m+m^2=(m-5)^2$ $\Rightarrow m=5$ $\Rightarrow n=-5-3\times 5=-20$ **ALT** $-3=\dfrac{n-("-5")}{m} \Rightarrow m=\dfrac{-(n+5)}{3}$ $2250=(50-m)^2+(-5-n)^2$ $2250=\left(50+\dfrac{n+5}{3}\right)^2+n^2+10n+25$ $2250=2500+\dfrac{100(n+5)}{3}+\dfrac{n^2+10n+25}{9}+n^2+10n+25$ $\Rightarrow \quad 0=9(2500-2250)+300(n+5)+10n^2+100n+250$ $=2500+300n+1500+10n^2+100n$ $=n^2+40n+400$ $=(n+20)(n+20)$ $\Rightarrow n=-20$ $\Rightarrow m=\dfrac{-(-20+5)}{3}=5$	M1 M1 M1 M1 A1 A1 [6] [M1] [M1] [M1] [M1] [A1] [A1]
(e)	Correctly uses the formula for a kite using their lengths	M1

Type MS · Using a line's equation to find a missing coordinate or intercept

Answer 4 of 7 · Question 10 part b · 1 mark

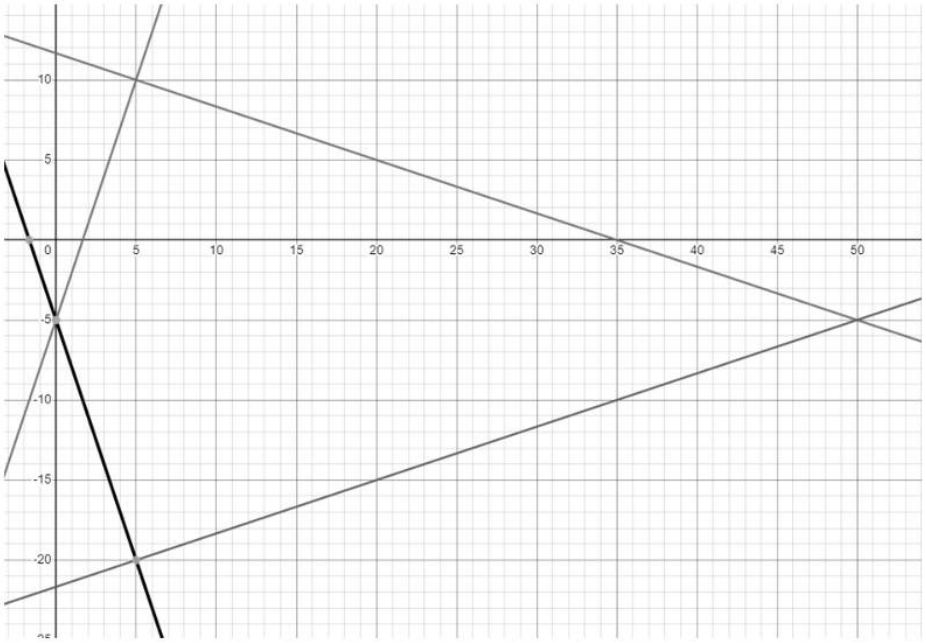
2025 NOV · Paper 01 · extract 2 of 4 · find part b in this mark scheme extract

	$\text{Area} = \frac{50\times(10-(-20))}{2} = 750$	dM1A1 [3]
	ALT 1 Correctly uses determinant method with their co-ordinates $\text{Area} = \frac{1}{2}\begin{pmatrix} 0 & 5 & 50 & 5 & 0 \\ -5 & 10 & -5 & -20 & -5 \end{pmatrix} = 750$	[M1 dM1A1]
	ALT 2 Correctly uses formulae for two triangles ABD and ACD using their lengths $\text{Area} = 2\times\left(\frac{1}{2}\times 15\times 50\right) = 750$ Or $\text{Area} = \left(\frac{1}{2}\times 30\times 5\right)+\left(\frac{1}{2}\times 45\times 30\right) = 750$	[M1 dM1A1]
		Total 17 marks

Type MS · Using a line's equation to find a missing coordinate or intercept

Answer 4 of 7 · Question 10 part b · 1 mark

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Type MS · Using a line's equation to find a missing coordinate or intercept

Answer 4 of 7 · Question 10 part b · 1 mark

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Question	Scheme	Marks

Type MS · Using a line's equation to find a missing coordinate or intercept

Answer 5 of 7 · Question 8 part c · 1 mark

2023 JUN · Paper 02 (regional R) · extract 1 of 5 · find part c in this mark scheme extract

Question	Scheme	Marks
8(a)	$\frac{y-5}{5-9}=\frac{x-1}{1-9}$ $\Rightarrow x-2y+9=0$	M1A1 A1 [3]
(b)	Coordinates of point X $\left(\frac{3\times9+1\times1}{3+1},\frac{3\times9+1\times5}{3+1}\right)=(7,8)$ The perpendicular gradient $=-\frac{1}{\frac{1}{2}}=-2$ Equation of l $y-8=-2(x-7)\Rightarrow y=-2x+22$ *	B1B1 B1ft M1A1 cso [5]
(c)	$y=-2(6)+22\Rightarrow p=10$	B1 [1]
(d)	$\overrightarrow{BA}=\begin{pmatrix}1-9\\5-9\end{pmatrix}=\begin{pmatrix}-8\\-4\end{pmatrix}\Rightarrow \overrightarrow{CD}=\begin{pmatrix}6-8\\10-4\end{pmatrix}=\begin{pmatrix}-2\\6\end{pmatrix}$ and coordinates of C are (6, 10) so coordinates of D are $(-2, 6)$	M1 A1A1 [3]
(e)	Length of AB (or CD) $=\sqrt{(9-5)^2+(9-1)^2}=\sqrt{80}$ Length of $CX=\sqrt{(10-8)^2+(7-6)^2}=\sqrt{5}$ Area of parallelogram $ABCD=\sqrt{5}\times\sqrt{80}=\sqrt{400}=20$ (units^2)	B1 B1 M1A1 [4]
		Total 16 marks

Part	Mark	Notes
(a)	M1	For using a correct method and the given coordinates of A and B to form the equation of AB. Do not score this mark until they find the gradient and form the equation of the line using a correct formula. If they use $y=mx+c$ do not allow this mark until they find c and form a complete equation. $\left[m=\frac{1}{2},\quad c=\frac{9}{2}\quad y=\frac{x}{2}+\frac{9}{2}\right]$
	A1	For the correct equation of AB in any form.
	A1	For the correct equation of AB in the required form.

Type MS · Using a line's equation to find a missing coordinate or intercept

Answer 5 of 7 · Question 8 part c · 1 mark

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(b)	B1	For either x or y correct coordinates of point X **NB This is an M mark in Epen**
	B1	For both x and y correct coordinates of point X **NB This is an A mark in Epen.**
	B1ft	For writing down the inverse reciprocal of their gradient for AB Ft their gradient form part (a)
	M1	For forming an equation for l using their coordinates of X and their negative reciprocal gradient of AB $y-8=-2(x-7)\Rightarrow y=-2x+22$
	A1	For the correct equation of l as shown.
(c)	B1	For $y = 10$
(d)	M1	For a suitable method. **Method 1 - Uses vectors:** $\overrightarrow{BA}=\begin{pmatrix}1-9\\5-9\end{pmatrix}=\begin{pmatrix}-8\\-4\end{pmatrix}\Rightarrow\overrightarrow{CD}=\begin{pmatrix}6+(-8)\\10+(-4)\end{pmatrix}\Rightarrow\left[\begin{pmatrix}-2\\6\end{pmatrix}\right]$ OR: $\overrightarrow{BC}=\begin{pmatrix}6-9\\9-10\end{pmatrix}=\begin{pmatrix}-3\\-1\end{pmatrix}\Rightarrow\overrightarrow{AD}=\begin{pmatrix}1-3\\5-(-1)\end{pmatrix}\Rightarrow\left[\begin{pmatrix}-2\\6\end{pmatrix}\right]$ **Method 2 - Uses simultaneous equations**: The equation of AD is $y=-\frac{x}{3}+\frac{16}{3}$ and of CD is $y=\frac{x}{2}+7$ $-\frac{x}{3}+\frac{16}{3}=\frac{x}{2}+7\Rightarrow x=-2$ and $y=6$ **Method 3 - Uses the gradient and length of *AB* or *CD*** $AB=CD=4\sqrt{5}=\sqrt{(6-x)^2+(y-10)^2}$ Gradient: $\frac{1}{2}=\frac{10-y}{6-x}\Rightarrow x=2y-14$ $\left(4\sqrt{5}\right)^2=(6-(2y-14))x^2+(y-10)^2\Rightarrow 5y^2-100y+420=0$ $\Rightarrow y=6,14$ and $x=-2,14$ Allow no more than one error in either method.
	A1	For either correct x or y coordinate of D [(– 2, 6)] **NB** this is an A mark in Epen
	A1	For both correct coordinates of D (– 2, 6)

Type MS · Using a line's equation to find a missing coordinate or intercept

Answer 5 of 7 · Question 8 part c · 1 mark

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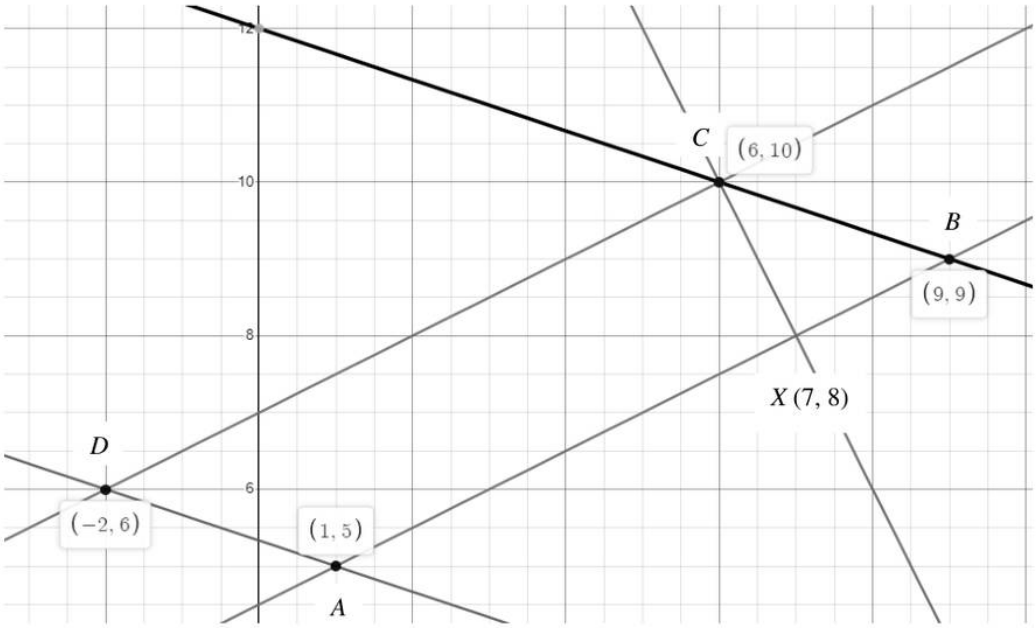
(e)	B1	For the correct length of either ***AB* [*CD*] or *CX*** These are given coordinates there is no ft
	B1	For the correct lengths of both ***AB* [*CD*] and *CX*** Ft their C and their X
	M1	For using a correct method to calculate the area of a parallelogram. $\text{b} \times \text{h} = CX \times AB = \sqrt{5} \times \sqrt{80} = \sqrt{400} = 20$ If their lengths are incorrect allow this mark provided their lengths are identified as base and perpendicular height. If they use AD as the base $\left[\sqrt{10}\right]$, the perpendicular height required is $2\sqrt{10}$ so base × height = $\sqrt{10} \times 2\sqrt{10} = ...$
	A1	For the correct area of 20 [units^2]
	ALT – Using determinants	
	B1B1	For sight of the correct array which must use the coordinates of A, B, C and D **ONLY.** The coordinates (7, 8) seen in the array is B0B0 $\begin{pmatrix} 1 & 9 & '6' & '-2' & 1 \\ 5 & 9 & '10' & '6' & 5 \end{pmatrix}$ Award both marks for fully correct. Award B1 if there are no more than 2 errors but with none missing. There must be 5 sets of coordinates in the array with first and last the same. **The coordinates must go in order around the parallelogram clockwise or anticlockwise.**
	M1	For the correct evaluation of their 2×5 array. Allow this even if they have (7, 8) included instead of (9, 9) [which is a common error] $\frac{1}{2}\begin{pmatrix} 1 & 9 & 6 & -2 & 1 \\ 5 & 9 & 10 & 6 & 5 \end{pmatrix} = \frac{1}{2}\left\lvert\left[(9+90+36-10)-(45+54-20+6)\right]\right\rvert = ...$
	A1	For the correct area of 20 [units^2]

Type MS · Using a line's equation to find a missing coordinate or intercept

Answer 5 of 7 · Question 8 part c · 1 mark

2023 JUN · Paper 02 (regional R) · extract 4 of 5 · find part c in this mark scheme extract

Useful sketch



Type MS · Using a line's equation to find a missing coordinate or intercept

Answer 5 of 7 · Question 8 part c · 1 mark

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Question	Scheme	Marks

Type MS · Using a line's equation to find a missing coordinate or intercept

Answer 6 of 7 · Question 4 part a · 3 marks

2023 NOV · Paper 01 · extract 1 of 2 · find part a in this mark scheme extract

Question	Scheme	Marks
4(a)	$(3\times14-2\times12-p=0\Rightarrow)\ p=18$	B1
	$3y-2q-"18"=0\Rightarrow 3\times2-2q-"18"=0\Rightarrow q=$ $q=-6$	M1 A1 [3]
(b)	$(X(x,y)=)\left(\frac{1\times"-6"+2\times12}{3},\frac{1\times2+2\times14}{3}\right)$ oe $=(6,10)$	B1B1 or M1A1
	$(\text{Gradient of } AB=)=\frac{2}{3}$ oe	B1
	$(\text{Gradient of } L=)-\frac{1}{"\frac{2}{3}"}\left(=-\frac{3}{2}\right)$	B1ft
	$y-"10"=-\frac{1}{"\frac{2}{3}"}(x-"6")\left(\Rightarrow y=-\frac{3}{2}x+19\right)$	M1
	$\Rightarrow 3x+2y-38=0$ oe Where a, b and c are integers.	A1 [6]
		Total 9 marks

	Mark	Notes
(a)	B1	For $p = 18$
	M1	Uses their value of p and substitutes $y = 2$ and finds a value for q.
	A1	For the correct value of q.
First 2 marks of (b) – use the scheme which gives the most marks to candidates.		
(b)	B1 (M1 ePen)	For one correct value (6, 10).
	B1 (A1 ePen)	For both correct values (6, 10).
	or M1	Correct method for finding both coordinates, using their q. Look for any equivalent methods eg similar triangles.
	or A1	Both correct values.
	B1	For a fully correct unsimplified or simplified gradient of AB Note this mark is awarded for **finding or stating** the gradient and if simplified, it must be clear that $\frac{2}{3}$ is not a fraction from the given ratio. Accept $\frac{14-2}{12--6}$ This mark may be implied for a correct gradient of $-\frac{3}{2}$ used in further work.
	B1ft	For finding the negative reciprocal of their gradient – does not need to be simplified.
	M1	For a full and correct method to find the equation for the line L using **their** coordinates of X and **their** perpendicular gradient. This need not be simplified. If $y = mx + c$ is used, the correct value of c must be found for their coordinates and gradient. It must be clear the candidate is using their coordinates for X and not, for example simply the numbers from the coordinates of A and B
	A1	For the correct equation of the line in the required form – any equivalent where a, b and c are integers.

Type MS · Using a line's equation to find a missing coordinate or intercept

Answer 6 of 7 · Question 4 part a · 3 marks

2023 NOV · Paper 01 · extract 2 of 2 · find part a in this mark scheme extract

Question	Scheme	Marks

Type MS · Using a line's equation to find a missing coordinate or intercept

Answer 7 of 7 · Question 5 part a · 2 marks

2023 JAN · Paper 02 (regional R) · extract 1 of 3 · find part a in this mark scheme extract

5(a)	$-\frac{1}{12}=\frac{10-0}{p-123}$ $\left[\text{or } -\frac{1}{12}=\frac{0-10}{123-p}\right]$ $\Rightarrow p-123=-12\times10\Rightarrow p=3^*$	M1 A1 cso [2]
(b)	$y-10=-\frac{1}{12}(x-3)\Rightarrow 12y+x-123=0$	M1A1 [2]
(c)	$m_k=12$ $y-10=12(x-3)\Rightarrow y=12x-26$	B1 M1A1 [3]
(d)	At C: $y=12x-26$ when $y=0$, $x=\frac{26}{12}$ oe $\text{Area}_{ABC}=\frac{1}{2}\times10\times\left(123-\frac{13}{6}\right)=\frac{3625}{6}$ ALT $\text{Area}=\frac{1}{2}\begin{vmatrix}3 & 123 & 13/6 & 3\\ 10 & 0 & 0 & 10\end{vmatrix}=\frac{3625}{6}$	M1A1 M1A1 [M1A1] [4]
		Total 11 marks

Type MS · Using a line's equation to find a missing coordinate or intercept

Answer 7 of 7 · Question 5 part a · 2 marks

2023 JAN · Paper 02 (regional R) · extract 2 of 3 · find part a in this mark scheme extract

Question	Notes	Marks
5(a)	States a correct expression for the gradient in terms of p either $-\frac{1}{12}=\frac{10-0}{p-123}$ or $-\frac{1}{12}=\frac{0-10}{123-p}$ and attempts to solve their equation in p $p-123=-12\times10\Rightarrow p=...$	M1
	Finds the value of $p=3$* Must show an intermediate step e.g. $k=p\pm c$ or $c=\frac{p}{k}$ or $c=kp$	A1 cso [2]
(b)	Forms an equation using either the formula or $y=mx+c$ with the given values $y-10=-\frac{1}{12}(x-3)$ OR $y-0=-\frac{1}{12}(x-123)$	M1
	For the correct equation in the required form $12y+x-123=0$ Coefficients must be integers. Allow for $r=1, s=12, t=-123$	A1 [2]
(c)	For the gradient of the normal when $x=3$ is 12	B1
	Forms an equation of the normal using either the formula or $y=mx+c$ with their values of the gradient of the normal. $y-12=12(x-3)$	M1
	For the correct equation in the required form. $y=12x-26$	A1 [3]
(d)	Attempts to find the x coordinate at point C $y=12x-26$ when $y=0$, $x=...$ Their linear equation from (c) with $y=0$ and attempt to solve.	M1
	For the correct value of x $x=\frac{26}{12}$ oe	A1
	For attempting to find the area of triangle ABC Area $=\frac{1}{2}\times10\times\left(123-\frac{13}{6}\right)=...$ or Area $=\frac{1}{2}\begin{vmatrix}3 & 123 & 13/6 & 3\\10 & 0 & 0 & 10\end{vmatrix}$	M1
	For the correct area of the triangle $\frac{3625}{6}$ oe isw rounding	A1 [4]
		Total 11 marks

Type MS · Using a line's equation to find a missing coordinate or intercept

Answer 7 of 7 · Question 5 part a · 2 marks

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Question	Scheme	Marks